OXFORD UNIVERSITY PRESS | Child Development

An associative learning account of how saliva becomes a cue for comfort

Journal:	*Child Development*
Manuscript ID	Draft
Manuscript type:	Review
Subject Area:	Learning < Cognitive Processes, Social cognition < Social, Emotional, Moral, Personality, Cognitive Processes, Conceptual development < Cognitive Processes
Keywords:	development, cognitive processes, social cognition
Abstract:	One challenge that infants face is learning who is related to whom. A recent study in Science by Thomas et al. (2022) showed that 8.5- to 10-month-olds use saliva sharing as an indicator of relationship closeness: infants expect individuals who previously shared saliva to comfort one another when distressed. The current paper offers a mechanistic explanation for this finding. We propose that through everyday experiences, saliva sharing and comforting become linked to a common representation that supports bidirectional expectations such that each behavior predicts the other. In a first simulation, we show that when this proposal is implemented in a connectionist computational model, the model uses prior saliva sharing as a predictor of comfort and accounts for infants' looking responses in Thomas et al. (2022). In a second simulation, we show that the model also uses prior comforting as a predictor of saliva sharing. In a third simulation, we show that this expectation increases with experience. The results of the second and third simulations generate novel predictions that can be tested in future infant studies. Taken together, our findings suggest that domain-general associative learning can explain how infants come to understand the social meaning of saliva sharing.

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"How saliva becomes a cue for comfort"

Abstract

One challenge that infants face is learning who is related to whom. A recent study in *Science* by Thomas et al. (2022) showed that 8.5- to 10-month-olds use saliva sharing as an indicator of relationship closeness: infants expect individuals who previously shared saliva to comfort one another when distressed. The current paper offers a mechanistic explanation for this finding. We propose that through everyday experiences, saliva sharing and comforting become linked to a common representation that supports bidirectional expectations such that each behavior predicts the other. In a first simulation, we show that when this proposal is implemented in a connectionist computational model, the model uses prior saliva sharing as a predictor of comfort and accounts for infants' looking responses in Thomas et al. (2022). In a second simulation, we show that the model also uses prior comforting as a predictor of saliva sharing. In a third simulation, we show that this expectation increases with experience. The results of the second and third simulations generate novel predictions that can be tested in future infant studies. Taken together, our findings suggest that domain-general associative learning can explain how infants come to understand the social meaning of saliva sharing.

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"How saliva becomes a cue for comfort"

To make sense of the social world, preverbal infants must learn to detect social relationships and evaluate their strength, or "thickness." Thick relationships—such as those among immediate family members—are marked by strong levels of attachment, loyalty, obligation, and mutual responsiveness (Thomas, 2024; Thomas et al., 2022). Distinguishing thick relationships from thinner ones enables infants to predict how people are likely to behave. For example, a father might be expected to comfort his own child, but not another child. However, despite the clear social utility of detecting thick relationships, the developmental origins of this ability are poorly understood.

One account proposes that infants infer the strength of social relationships by observing external cues, and that saliva sharing may serve as a particularly reliable predictor of relationship thickness. Thick relationships are characterized by behaviors such as emotional responsiveness, physical affection, comforting, and cooperation in daily tasks. However, each of these behaviors can also occur within thinner relationships, making them less individually predictive of relationship thickness. In contrast, saliva sharing—through shared food, drinks, and utensils—occurs frequently in thick relationships but rarely in thin ones. Moreover, unlike culturally specific gestures that can signal closeness (e.g., bowing, cheek-kissing), saliva sharing appears in every known culture (Thomas et al., 2022). Thus, saliva sharing may be a reliable, universal indicator of thick relationships.

A recent study published in *Science* by Thomas et al. (2022) tested whether infants treat saliva sharing as a cue for relationship thickness by testing whether infants expect individuals who previously shared saliva to comfort one another when distressed. Although Thomas et al. (2022) also tested older children, we focus on infants who cannot rely on explicit instruction. In

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Thomas et al.'s primary experiment (i.e., Exp. 2B), 8.5- to 10-month-old infants saw a puppet interact with two actresses. The puppet shared saliva with the first actress by eating from the same orange slice. The puppet shared a ball with the second actress but did not share saliva. Infants then saw the same puppet positioned between the two actresses on the screen. The puppet exhibited distress by crying. As the puppet cried, both actresses turned towards the puppet. Thomas et al. (2022) reasoned that if infants expected saliva-sharers to comfort one another when distressed, infants should look first and longest at the actress who ate from the same orange slice as the puppet—and this is exactly what they found.

To test whether infants' expectations were specific to previous saliva-sharing partners, rather than a general expectation that saliva-sharers comfort *anyone* in distress, Thomas et al. (2022) repeated the initial study but replaced the original puppet with a new puppet at the beginning of the distress event (Experiment 2D). When the new puppet was in distress, infants looked equally long at both actresses. The findings from these two studies suggest that infants expect saliva-sharers to comfort their saliva-sharing partners but not distressed individuals generally.

In a third experiment (Experiment 3B), Thomas et al. (2022) tested whether infants' behavior in the previous experiments was specifically due to saliva sharing (rather than food sharing). They showed infants a video of a central actress flanked by two puppets. First, the actress placed her finger into her mouth, rotated it, and then placed it into one of the puppet's mouths. Next, the actress placed her finger on her forehead, rotated it, and placed it on the other puppet's forehead. The actress then expressed distress, and Thomas et al. (2022) measured which puppet infants looked at first and longest. Although infants' first looks were split evenly between both puppets, infants looked longer at the puppet with whom the actress shared saliva. Thomas et

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al. (2022) interpreted these findings—which were partially replicated by Ciftci et al. (2025)—to mean that preverbal infants specifically use saliva sharing to predict who should provide comfort to whom.

An associative learning account for how saliva sharing comes to indicate comfort

Although Thomas et al. (2022) suggest that there may be an evolutionary basis for the expectation that saliva-sharers comfort one another, they remain largely agnostic about its developmental origins. Here, our goal is to provide a mechanistic account of how saliva sharing comes to predict comfort. Our central claim is that infants' expectations about thick relationships are the product of their experiences. We propose that a domain-general associative learning mechanism links saliva sharing and comforting to a common representation that supports bidirectional expectations such that each behavior predicts the other. Put simply, infants learn from experience that the same individuals tend to share saliva and comfort one another during times of distress.

On our account, as infants repeatedly witness others sharing saliva, they gradually build a representation of saliva-sharing events. Here, "representation" refers to a pattern of activation across units in an associative, cognitive system that are elicited by different environmental events. Infants' representation of saliva-sharing events will include information about who shared saliva with whom. Importantly, because the specific food items may vary from one saliva-sharing event to the next, the association between the act of saliva sharing and the representation that is evoked by the act of saliva sharing will be stronger than the association between any one food item and this same representation. As infants accrue experience with individuals comforting

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one another during moments of distress, they establish and build a parallel representation of those events.

If the same individuals who share saliva also provide comfort other time points, these separate representations of saliva sharing and comforting can merge to form a common representation. Specifically, when infants see a former saliva sharer providing comfort, the representation of saliva sharing is reactivated. While this representation is active, new information (i.e., comforting) is incorporated into it. As a result, a common representation emerges that contains information about saliva sharing and comforting, even though saliva sharing and comforting were not experienced simultaneously. A corollary of such a common representation is that infants' expectations about saliva sharing and comforting actions should be bidirectional such that saliva sharing predicts comfort, and comfort predicts saliva sharing. Note that our account assumes that infants can track the identity of individuals across space and time. Because prior work shows that that this capacity emerges between 6 and 12 months of age (Káldy & Leslie, 2003, 2005; Kibbe & Leslie, 2011, 2013, 2016), and the infants tested in Thomas et al. (2022) fall within this range, we contend that this is a reasonable assumption.

In the next section, we present a connectionist computational model that instantiates this account. Across three simulations, we show that our model explains infants' looking responses in Thomas et al. (2022) (Simulation 1) and implies that infants may hold bidirectional expectations about the link between saliva sharing and comfort (Simulation 2) that grow in strength with experience (Simulation 3).

All simulations were run using the Lens neural network simulator (Rohde, 1999). All training and testing files, including the data from the simulations, are available for download at https://osf.io/9mqyn/overview?view_only=c09f319c3fa643d9b688cb09e0794379.

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General Method

Network Architecture

The simulations reported here used a three-layer, feedforward, autoassociative neural network model (e.g., Mareschal et al., 2000) trained with backpropagation in cross-entropy error (e.g., Rumelhart et al., 1986). The hidden and output units used a sigmoid activation function. Gaussian noise was added to input activations to reflect infants' noisy sensory processing (M = 0, SD = 0.875).

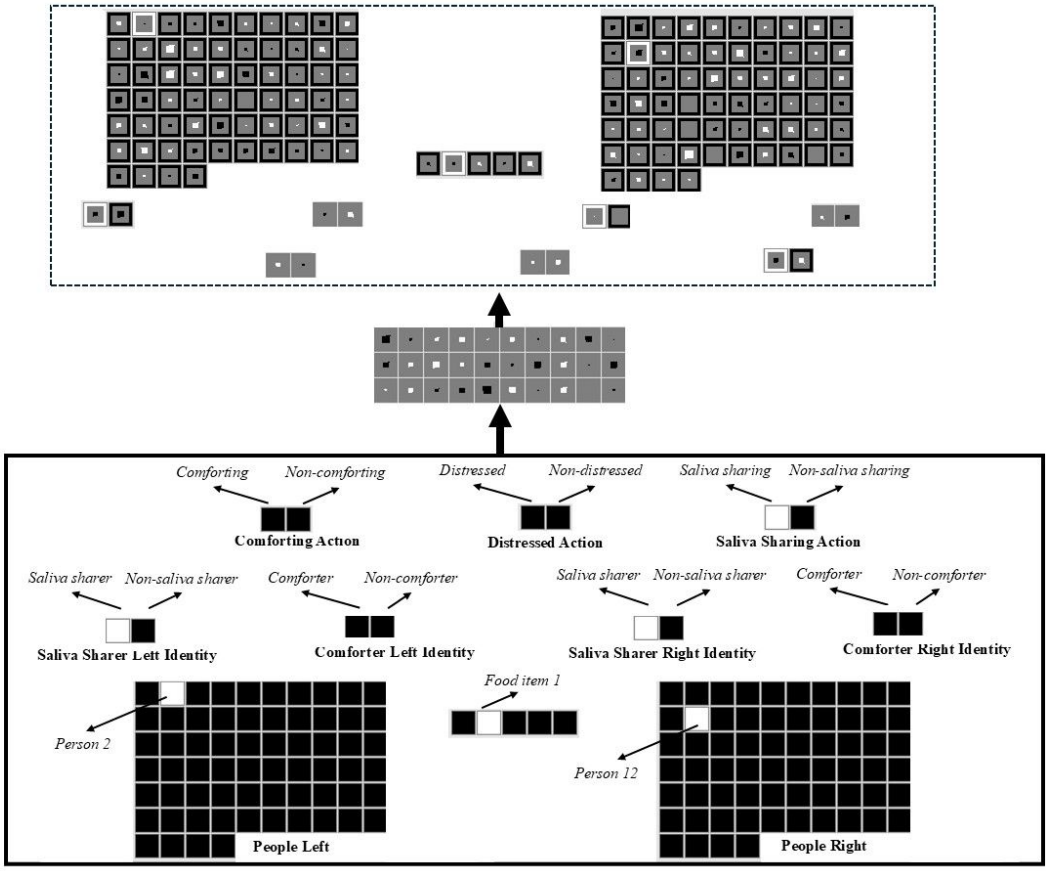

Figure 1. Schematic of an untrained model architecture used in the current simulations. The input and output groups are identical. The input groups are written in bold. The model consisted of 10 structurally identical input and output groups: People Left/Right, Saliva Sharing Left/Right Identity, Comforting Left/Right Identity, Comforting Action, Distressed Action, Food Item, and Saliva Sharing Action. The specific feature that a given unit coded for is shown in italics within each group.

The model consisted of three layers (Figure 1). There were 147 input units, 30 hidden units, and 147 output units. The 147 input and output units were divided into 10 groups of units—the output groups were identical to the input groups. These groups—along with the number of units that comprised them—are shown in Figure 1. Saliva Sharing Identity and Comforting Identity groups captured the *identity* of being a saliva sharer or comforter; the Saliva Sharing Action and Comforting Action groups captured the *action* of engaging in saliva sharing or comforting. By separating an individual's identity from its actions, it was possible for the model to represent situations in which a former comforter engages in saliva sharing or a former saliva sharer engages in comforting. For each input group, the model's task was to activate in the output layer exactly those units that were active in the input layer. Localist representations—in which a single unit represented a single concept—were adopted for each group to simplify the interpretation and analysis of the model. Although the names of the groups and features were conceptually laden (e.g., Distressed Action), we do not intend to suggest that infants see these features in a semantically meaningful or rich way. We use these terms simply to make it easier to understand the structure of the model. Finally, because saliva/non-saliva sharing and comforting/non-comforting necessarily occur between two people, it was always the case in the current simulation that the model saw two people (i.e., at all points a single unit was turned on in the People Left/Right groups).

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Pretraining

Table 1. Eight pretraining events used to approximate infants' everyday experience with saliva sharing, comforting, and distress, including events that link these actions.

Event Type	Key Action(s)	Purpose of Pretraining

Saliva sharers + saliva sharing	Two individuals sharing saliva via different foods	Encode representation of saliva sharing
Saliva sharers + comfort + distress	Former saliva sharer comforts their saliva-sharing partner when distressed	Establish a common representation linking saliva sharing and comfort via shared individual
Non-saliva sharers + non-saliva sharing	Two individuals share something other than saliva	Encode representation of non-saliva sharing
Non-Saliva sharers + non-Comfort + Distress	One individual does not comfort their non-saliva-sharing partner when distressed	Establish a common representation linking non-saliva sharing and non-comfort via shared individual
Comforters + comforting + distress	One individual comforts their distressed partner	Encode representation of comforting
Comforters + saliva sharing	A former comforter shares saliva with their previously distressed partner via different foods	Establish common representation linking comforting and saliva sharing via a shared individual
Non-Comforter + non-comforting + distress	One individual fails to comfort a distressed partner	Encode representation of non-comforting
Non-comforter + non-saliva sharing	One individual does not share saliva does not provide comfort	Establish common representation linking non-comforting and non-saliva sharing via a shared individual

The pretraining phase—which approximated infants' real-world experience—lasted from anywhere between 600 and 800 epochs, in increments of 100. Different numbers of epochs were

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used to ensure that the network's performance was not idiosyncratic to the length of the pretraining phase, and different food items were used in each saliva sharing event to instantiate the idea that the crucial link is between saliva sharing and comforting rather than between food sharing and comforting. Networks experienced eight kinds of events during training (Table 1). A unique combination of people was shown in each instance. These events instantiated our assumptions about the kinds of real-world, saliva sharing and comforting experience that influenced infants' looking behaviors in Thomas et al. (2022). Each event was implemented by activating the relevant units in the Person, Saliva Sharing/Comforting Identity groups, Distressed Action, Saliva Sharing/Comforting Action groups (see the Appendix for details about which features were turned on for each event type).

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Familiarization

Following the pretraining phase, networks were randomly assigned to 1 of 3 conditions (20 networks per condition). The conditions corresponded, respectively, to Experiments 2B, 2D, and 3B in Thomas et al. (2022). A "participant" corresponded to an untrained model whose weights were initialized to small, random values (sampled uniformly between ±0.1). Note that Condition 3 was identical to Conditions 1 and 2 except that food information was missing (as was the case in Thomas et al., 2022). The familiarization phase for networks in all three conditions involved an event in which food sharing occurred between a central character and another individual and an event in which non-food sharing occurred between the same central character and a second individual. Because our theory does not specify whether the critical link between saliva sharing and comfort depends on who shared versus received saliva, we did not model the direction of sharing. This explains why we describe the events in general terms rather than in terms of who shared and received saliva. All events used novel people and were

implemented in a similar manner to the pretraining events (see the Appendix). Finally, the familiarization phase could last for either 5 or 10 epochs. Different values ensured that the model's performance was robust to the length of the familiarization phase.

Test

Following the familiarization phase, networks participated in a test phase. The nature of this test phase depended on the condition to which the networks were assigned. Networks in Conditions 1 and 3 were shown an event in which the same central character expressed distress. In contrast, networks in Condition 2 were shown the same event except that the character that was in distress was different from that shown during the familiarization phase.

The dependent measure used in the present simulations was cross-entropy error. Typically, greater error corresponds to longer looking. However, because the looking times in Thomas et al. (2022) were reported in terms of expectations rather than surprise, we applied a simple numerical transformation to make the present error values more comparable to those data. Specifically, we subtracted a given network's error score produced by a test event from the maximum error for that test event. This transformation means that lower network error corresponded to longer looking or greater expectation.

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Results

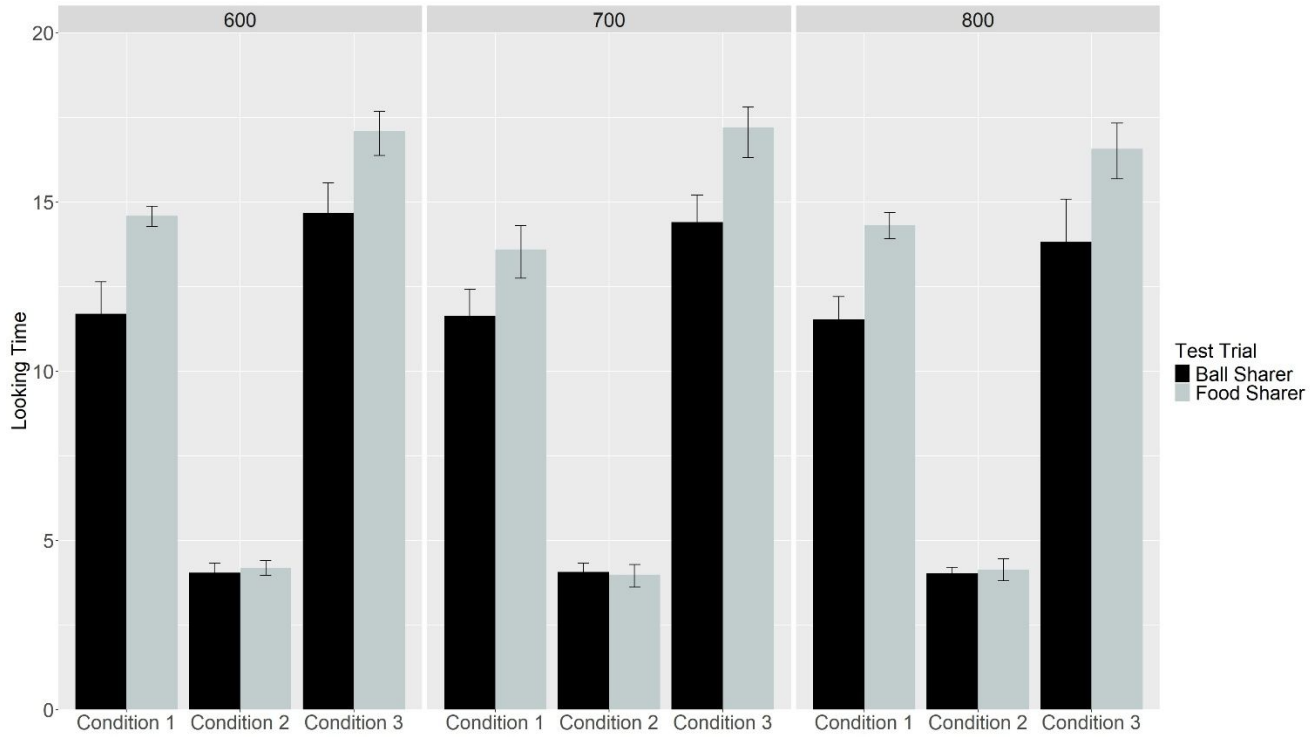

Figure 2. Networks' mean looking time to the Ball Sharer and Food Sharer test events across Conditions and Pretraining Epochs.

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Figure 2 shows networks' looking times to the Food and Ball Sharers across Conditions 1-3 and Pretraining Epochs. Data were entered into a linear mixed-effects model with Test Trial as the within-subjects fixed effect (Ball Sharer vs. Food Sharer), Condition (Condition 1 vs. Condition 2 vs. Condition 3) and Pretraining Epochs (600 vs. 700 vs. 800) as between-subjects fixed effects, and participants as the random effect. Familiarization Epoch (or the number of familiarization trials) was not included as an additional effect because it was colinear with Pretraining Epochs and created singularity issues for the model. The results revealed main effects of Test Trial and Condition, which were qualified by an interaction between the two, $F(2, 351) = 24.73$, $p < .001$. Follow-up analyses were conducted for each Condition to assess the nature of this interaction. A paired t-test for Conditions 1 and 3 showed that networks looked longer at the Food Sharer (Condition 1: $M = 14.16$, $SD = 1.80$; Condition 3: $M = 16.95$, $SD = 2.43$) than at the

Ball Sharer (Condition 1: $M = 11.61$, $SD = 2.83$; Condition 3: $M = 14.29$, $SD = 3.53$), both t's > 6.84, both p's < .001. In contrast, networks in Condition 2 looked equally long at the Food Sharer ($M = 4.10$, $SD = 0.99$) and Ball Sharer ($M = 4.05$, $SD = 0.85$) test events, $t(119) = 0.44$, $p = .66$.

Discussion

Simulation 1 tested whether infants' looking behaviors in Thomas et al. (2022) could be explained by a domain-general learning mechanism that links the acts of saliva sharing and comforting to a shared representation, particularly when those experiences involve the *same* individuals. The results indicated that infants' responses could be explained by such a mechanism. Because a shared representation would also support bidirectional expectations about the link between saliva sharing and comfort, in Simulation 2, we tested whether comfort also predicts saliva sharing.

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Simulation 2

Network Architecture and Familiarization

The architecture used in Simulation 2 was identical to that in Simulation 1. The pretraining phase was also identical.

Familiarization and Test

Following pretraining, networks (N = 20) were familiarized to an event in which the central character was comforted by one of two individuals, but not the other. Networks in the experimental condition were then tested on an event in which the central character from the familiarization phase engaged in saliva sharing with the former comforter and non-comforter. A

separate group of networks (N = 20) in the Control condition were familiarized to the same event as the experimental group and tested on an identical event except that the central character was replaced with a new one.

Results

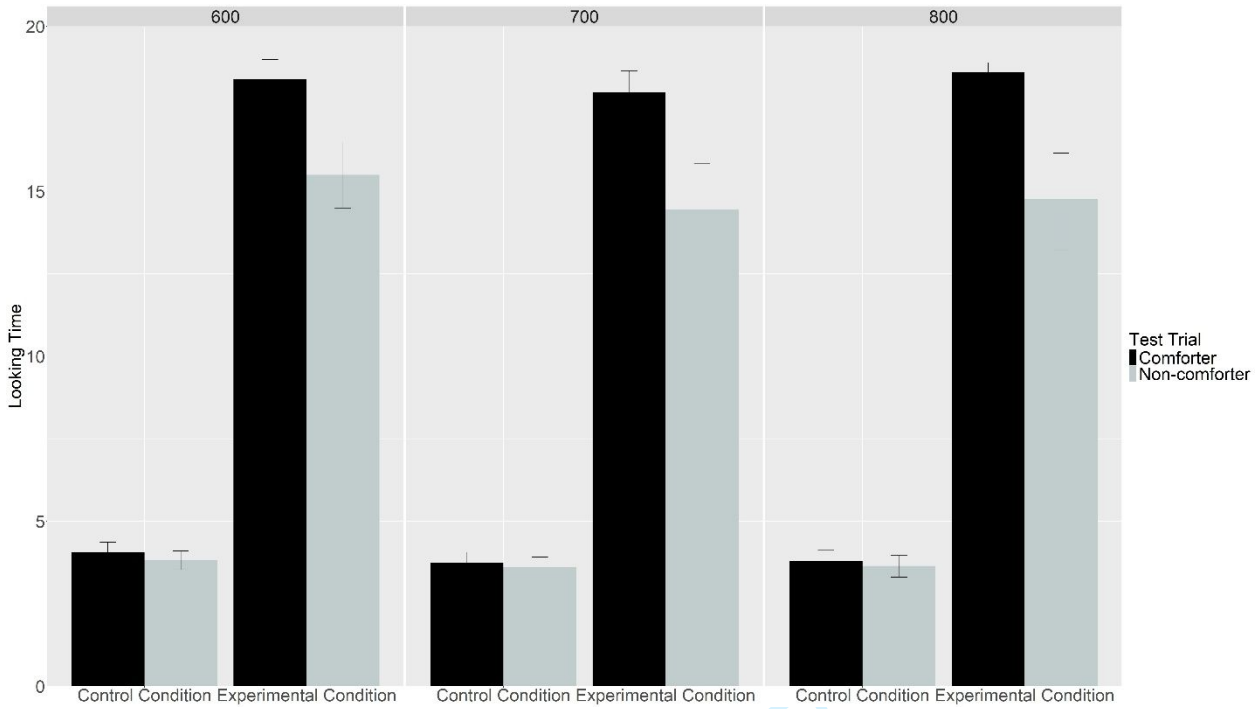

Figure 3. Networks' mean expectation or looking time to the Comforter and Non-Comforter test events across Conditions and Pretraining Epochs.

Figure 3 shows networks' looking times to the Comforter and Non-Comforter across the Experimental and Control Conditions and Pretraining Epochs. Data for networks in these conditions were entered into a linear mixed-effects model with Test Trial as the within-subjects fixed effect (Comforter vs. Non-Comforter), Condition (Experimental vs. Control) and Pretraining Epoch (600 vs. 700 vs. 800) as the between-subjects fixed effects, and participants as the random effect. The results revealed main effects for Condition and Test trial, both of which

were qualified by an interaction between Condition and Test Trial, $F(1, 234) = 51.16$, $p = 0$. Follow-up analyses were conducted for each Condition. A paired t-test for the Experimental Condition showed that networks looked longer at the Comforter ($M = 18.33$, $SD = 1.94$) than the Non-comforter ($M = 14.90$, $SD = 4.34$), $t(119) = 7.73$, $p = 0$. In contrast, networks in the Control Condition looked equally long at the Comforter ($M = 3.86$, $SD = 1.06$) and Non-comforter ($M = 3.69$, $SD = 1.02$), $t(119) = 1.69$, $p = .09$.

Discussion

Simulation 2 demonstrates that a common representation can support bidirectional expectations about saliva-sharing and comforting and generates a novel, testable prediction for future behavioral research. However, an evolved predisposition to interpret both actions as cues to relationship thickness could also give rise to bidirectional expectations. Thus, behavioral data confirming this prediction would not distinguish between associative learning and evolutionary accounts. The two accounts, however, make different developmental predictions: if bidirectional expectations arise from associative learning, their strength should increase with experience; if they reflect an evolved predisposition, their strength should remain relatively stable across development. Simulation 3 tested whether such expectations strengthen through associative learning.

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Simulation 3

Network Architecture, Familiarization, and Test

Simulation 3 was identical to Simulation 2 except that networks were tested at three different "ages": young, intermediate, and old. The pretraining phase lasted for 100 epochs for

young networks, 300 epochs for intermediate networks, and 800 epochs for old networks. This parameter implemented the assumption that infants at different ages necessarily enter experiments with differing amounts of real-world experience. Additionally, the learning rate parameter for young networks was 0.001, 0.01 for intermediate networks, and 0.1 for old networks. This parameter controls the magnitude of the weight changes in the model and corresponds theoretically to the notion that processing speed improves with age. Finally, the number of hidden units was 10 for young networks, 20 for intermediate networks, and 30 for old networks. This parameter captured the idea that infants' processing capacity improves with age. All three parameters implemented a simple model of development that is consistent with existing developmental simulation studies (e.g., Benton, 2024; Rakison & Lupyan, 2008).

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Results

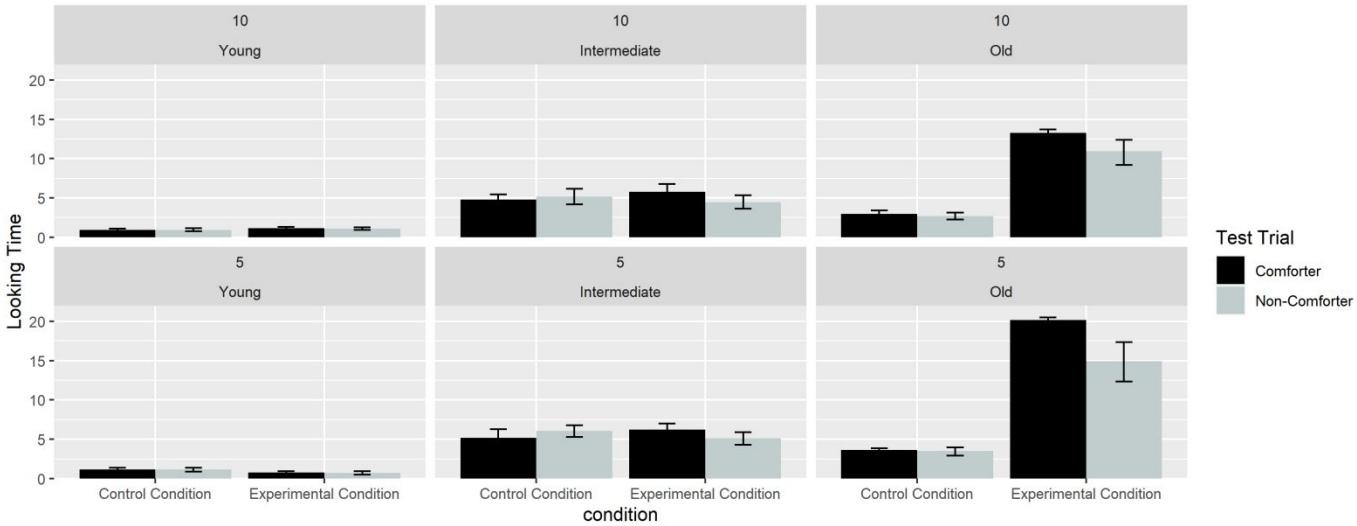

Figure 4. Networks' mean looking time to the Comforter and Non-Comforter across Conditions, Age, and Familiarization Epochs.

Figure 4 shows networks' looking times to the Comforter and Non-Comforter across the Experimental and Control Conditions, Age, and Familiarization Epochs. Data for networks in

these conditions were entered into a linear mixed-effects model with Test Trial as the within-subjects fixed effect (Comforter vs. Non-Comforter), Condition (Experimental vs. Control), Familiarization Epoch (5 vs. 10), and Age (Young vs. Intermediate vs. Old) as the between-subjects fixed effects, and participants as the random effect. The results revealed several main effects and two-way interactions, which were qualified by two three-way interactions. The first three-way interaction was between Test Trial, Age, and Condition, $F(2, 228) = 8.97$, $p < .001$. The second three-way interaction was between Age, Condition, and Familiarization Epoch, $F(2, 228) = 20.74$, $p = 0$. Here, we only focus on the first three-way interaction because the second one was not theoretically meaningful. Follow-up analysis for each Age and Condition showed that the young networks equally long at the Comforter and Non-Comforter in the Experimental and Control conditions, both t's < 0.77, both p's > .45. In contrast, intermediate networks looked longer at the Comforter ($M = 6$, $SD = 2.07$) than at the Non-Comforter ($M = 4.79$, $SD = 1.95$) in the Experimental condition, $t(39) = 2.61$, $p = .01$, but equally long at the Comforter ($M = 4.96$, $SD = 2.10$) and the Non-Comforter ($M = 5.61$, $SD = 2.13$) in the Control condition, $t(39) = -1.59$, $p = .12$. Similarly, old networks looked longer at the Comforter ($M = 16.71$, $SD = 3.62$) than at the Non-Comforter ($M = 12.92$, $SD = 5.34$) in the Experimental condition, $t(39) = 4.67$, $p = 0$, but equally long at the Comforter ($M = 3.28$, $SD = 0.95$) and the Non-Comforter ($M = 3.11$, $SD = 1.20$) in the Control condition, $t(39) = 0.75$, $p = .46$. Crucially, it was the case that the difference in looking time to the Comforter and Non-Comforter in the Experimental condition increased with age; that is, the difference in looking time to the Comforter and Non-Comforter was greater for the intermediate networks ($M = 1.21$, SD = 2.94) than for the young networks ($M = 0.05$, $SD = 0.38$), $t(40.29) = -2.49$, $p = .02$. Similarly, this difference was greater for the old networks ($M =$

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3.79, *SD* = 5.13) than the intermediate ones, $t(62.09) = -2.76$, $p < .01$. This supports the contention that the strength of the bidirectional expectation gets stronger with age.

Discussion

Simulation 3 showed that expectations of saliva sharing following comforting strengthened over time. Because an evolutionary account predicts stable expectations, this finding offers a novel testable prediction for behavioral studies that could distinguish between associative learning and evolutionary accounts.

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General Discussion

A recent study by Thomas et al. (2022) in *Science* showed that 8.5- to 10-month-old infants expect saliva-sharers to comfort each other during moments of distress. The goal of the current paper was to offer a mechanistic explanation for these findings and to test whether a computational instantiation of this account could explain them. Simulation 1 showed that an associative learning account that links saliva sharing to comforting via a common representation was sufficient to explain infants' looking behavior in Thomas et al. (2022). Simulation 2 extended this simulation by showing that expectations of saliva sharing and comforting could be bidirectional. Simulation 3 showed that an associative learning account predicts that the strength of these expectations will increase over time. Taken together, these results provide a mechanistic explanation of how preverbal infants learn to use saliva as a cue for comfort.

The current findings add to a broader tradition in the developmental cognitive sciences that use connectionist models to demonstrate the sufficiency of domain-general processes for capturing infants' behavior in tasks that rely on looking time (e.g., Benton, 2024, Benton & Lapan, 2021; Christiansen & Curtin, 1999; Cohen et al., 2002; Munakata et al., 1997;

Schlesinger & Casey, 2003; Vong et al., 2024). At the same time, the current simulations extend this extensive body of evidence by demonstrating that the basis of infants' expectation of comfort following saliva sharing may be associative learning.

Although our model demonstrates that associative learning is sufficient to explain how infants come to see saliva sharing as having social significance, it does not rule out the possibility—as suggested by Thomas et al. (2022)—that this expectation has an evolutionary basis. Additionally, the current simulations do not show that infants' knowledge about the social meaning of saliva sharing necessarily derives from associative processes. Addressing both questions will require behavioral experiments. For example, if infants use saliva sharing as a predictor of comforting and this expectation increases in strength over time, this would suggest that infants rely on associative learning to detect relationship thickness. A promising avenue for future research will be to test these predictions directly and to pit the current proposal against alternatives. Beyond these issues, our model could be critiqued by its use of localist, semantically neutral input representations. Such representations—which are neither rich nor multimodal—likely poorly approximate infants' real-world experience with saliva sharing and comfort. Future modeling work should instantiate the current proposal into neural networks with richer representations that better capture infants' natural experiences.

In sum, by demonstrating that a shared mental representation that arises from associative learning is sufficient to explain infants' responses in Thomas et al. (2022), we show how abstract social expectations can emerge from simple learning mechanisms and experience. More broadly, these findings underscore the value of computational modeling for elucidating potential mechanisms of early social reasoning.

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